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Hopefully running this is super simple. I program in an IDE and was unable to test this is a more traditional setting but here are the needed lines for testing.

This project was implemented in Python 3. No compilation step is required.

To run the server:

1. Open a terminal.
2. Navigate to the project directory.
3. Run: python server.py

The server will start listening on: http://localhost:8080

Accessing the website:

200 OK: http://localhost:8080/index.html
This will take you to the main page. You should see the back of a Magic: The Gathering card and some text. When accessing this page it creates a cookie as well with value = hello

24	3.473371000	127.0.0.1	127.0.0.1	TCP	56	8080 → 63247 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=65495 WS=256 SACK_PERM
25	3.473394900	127.0.0.1	127.0.0.1	TCP	44	63247 → 8080 [ACK] Seq=1 Ack=1 Win=65280 Len=0
26	3.473900000	127.0.0.1	127.0.0.1	HTTP	742	GET /index.html HTTP/1.1
27	3.473932800	127.0.0.1	127.0.0.1	TCP	44	8080 → 63247 [ACK] Seq=1 Ack=699 Win=64768 Len=0
28	3.474881400	127.0.0.1	127.0.0.1	TCP	145	8080 → 63247 [PSH, ACK] Seq=1 Ack=699 Win=64768 Len=101 [TCP PDU reassembled in 30]
29	3.474903300	127.0.0.1	127.0.0.1	TCP	44	63247 → 8080 [ACK] Seq=699 Ack=102 Win=65280 Len=0
30	3.474915900	127.0.0.1	127.0.0.1	HTTP	279	HTTP/1.0 200 OK (text/html)
31	3.474943000	127.0.0.1	127.0.0.1	TCP	44	63247 → 8080 [ACK] Seq=699 Ack=337 Win=65024 Len=0
32	3.474957400	127.0.0.1	127.0.0.1	TCP	44	8080 → 63247 [FIN, ACK] Seq=337 Ack=699 Win=64768 Len=0

Cookie:

301 Moved Permanently: http://localhost:8080/page1.html
This should automatically redirect you to page2.html and show text specifying that you've been redirected

484	102.191309400	::1	::1	TCP	64	8080 → 64221 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
485	102.191610700	::1	::1	TCP	76	49803 → 8080 [SYN] Seq=0 Win=65535 Len=0 MSS=65475 WS=256 SACK_PERM
486	102.191622100	::1	::1	TCP	64	8080 → 49803 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
487	102.495817700	127.0.0.1	127.0.0.1	TCP	56	59428 → 8080 [SYN] Seq=0 Win=65535 Len=0 MSS=65495 WS=256 SACK_PERM
488	102.495930700	127.0.0.1	127.0.0.1	TCP	56	8080 → 59428 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=65495 WS=256 SACK_PERM
489	102.495971800	127.0.0.1	127.0.0.1	TCP	44	59428 → 8080 [ACK] Seq=1 Ack=1 Win=65280 Len=0
490	102.496333700	127.0.0.1	127.0.0.1	TCP	56	50647 → 8080 [SYN] Seq=0 Win=65535 Len=0 MSS=65495 WS=256 SACK_PERM
491	102.496470100	127.0.0.1	127.0.0.1	TCP	56	8080 → 50647 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=65495 WS=256 SACK_PERM
492	102.496482900	127.0.0.1	127.0.0.1	TCP	44	50647 → 8080 [ACK] Seq=1 Ack=1 Win=65280 Len=0
527	111.324911300	127.0.0.1	127.0.0.1	HTTP	742	GET /page2.html HTTP/1.1
528	111.324941000	127.0.0.1	127.0.0.1	TCP	44	8080 → 59428 [ACK] Seq=1 Ack=699 Win=64768 Len=0
529	111.325521600	127.0.0.1	127.0.0.1	TCP	145	8080 → 59428 [PSH, ACK] Seq=1 Ack=699 Win=64768 Len=101 [TCP PDU reassembled in 531]
530	111.325543600	127.0.0.1	127.0.0.1	TCP	44	59428 → 8080 [ACK] Seq=699 Ack=102 Win=65280 Len=0
531	111.325557700	127.0.0.1	127.0.0.1	HTTP	187	HTTP/1.0 200 OK (text/html)
532	111.325565200	127.0.0.1	127.0.0.1	TCP	44	59428 → 8080 [ACK] Seq=699 Ack=245 Win=65280 Len=0
533	111.325581500	127.0.0.1	127.0.0.1	TCP	44	8080 → 59428 [FIN, ACK] Seq=245 Ack=699 Win=64768 Len=0
534	111.325589600	127.0.0.1	127.0.0.1	TCP	44	59428 → 8080 [ACK] Seq=699 Ack=246 Win=65280 Len=0
535	111.326239800	127.0.0.1	127.0.0.1	TCP	44	59428 → 8080 [FIN, ACK] Seq=699 Ack=246 Win=65280 Len=0

404 Error not found: http://localhost:8080/doesnotexist.html
This displays the classic 404 not found page.

535	111.326239800	127.0.0.1	127.0.0.1	TCP	44	59428 → 8080 [FIN, ACK] Seq=699 Ack=246 Win=65280 Len=0
536	111.326271400	127.0.0.1	127.0.0.1	TCP	44	8080 → 59428 [ACK] Seq=246 Ack=700 Win=64768 Len=0
665	145.738447000	127.0.0.1	127.0.0.1	HTTP	749	GET /doesnotexist.html HTTP/1.1
666	145.738476300	127.0.0.1	127.0.0.1	TCP	44	8080 → 50647 [ACK] Seq=1 Ack=706 Win=64768 Len=0
667	145.739101100	127.0.0.1	127.0.0.1	TCP	116	8080 → 50647 [PSH, ACK] Seq=1 Ack=706 Win=64768 Len=72 [TCP PDU reassembled in 669]
668	145.739120200	127.0.0.1	127.0.0.1	TCP	44	50647 → 8080 [ACK] Seq=706 Ack=73 Win=65280 Len=0
669	145.739133400	127.0.0.1	127.0.0.1	HTTP	219	HTTP/1.0 404 Not Found (text/html)
670	145.739141100	127.0.0.1	127.0.0.1	TCP	44	50647 → 8080 [ACK] Seq=706 Ack=248 Win=65280 Len=0
671	145.739157400	127.0.0.1	127.0.0.1	TCP	44	8080 → 50647 [FIN, ACK] Seq=248 Ack=706 Win=64768 Len=0
672	145.739166000	127.0.0.1	127.0.0.1	TCP	44	50647 → 8080 [ACK] Seq=706 Ack=249 Win=65280 Len=0
673	145.739504400	127.0.0.1	127.0.0.1	TCP	44	50647 → 8080 [FIN, ACK] Seq=706 Ack=249 Win=65280 Len=0
674	145.739534400	127.0.0.1	127.0.0.1	TCP	44	8080 → 50647 [ACK] Seq=249 Ack=707 Win=64768 Len=0
675	145.741304200	::1	::1	TCP	76	58391 → 8080 [SYN] Seq=0 Win=65535 Len=0 MSS=65475 WS=256 SACK_PERM

Assumptions:

- All requested files are stored in the same directory as the server program.
- The maximum HTTP request size is assumed to be 1024 bytes.
- Only basic HTTP 1.0 functionality is implemented.

Limitations:

- The server only supports the HTTP GET method
- Only the specified file types are supported
- The server does NOT implement persistent connections